Krishiv Singhal

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Education

Carnegie Mellon University *Aug 2024–Dec 2027*

Major in Mathematics and Logic and Computation *United States*

- Notable coursework: Advanced NLP (PhD), Machine Learning (PhD), Cooperative AI (Graduate), Modal Logic (Graduate), Formal Logic, Probability Theory, Linear Algebra, Abstract Algebra, Real and Complex Analysis. GPA: 3.7.
- Studied abroad at the University of Sydney; completed advanced coursework in maths on a full merit scholarship; awarded the Dalyell Scholarship and Chancellor's Award. GPA: Distinction.

United World College, Singapore *Aug 2017–May 2024*

International Baccalaureate *Singapore*

- Perfect IB Score: 45/45 (top 0.5% worldwide); ranked first in class for thesis on the Curry–Howard correspondence and intuitionism.

Research Experience

AI Evaluation Research Group *Sep 2026–Present*

Carnegie Mellon University

- Selected to Prof Sarah Cen's research group on prediction and recalibration of model performance under covariate shift, including synthetic distribution-shift experiments and unlabeled recalibration methods.

Probabilistic Modal Logic and Formal Verification *Mar 2026–Present*

Research, Carnegie Mellon University

- Developing graded probabilistic semantics for propositional dynamic logic, deriving quantitative interpretations of classical modal axioms over stochastic transition systems and exploring applications to formal safety specifications for AI agents. Research under Prof. Adam Bjorndahl.

Computational Linguistics Research Group *Sep 2024–Feb 2025*

Carnegie Mellon University

- Analyzed human and AI writing through spectral gaps of part-of-speech Markov chains; paired differences were significant in every genre–model comparison, strongest at $p \approx 5 \times 10^{-113}$.

Industry Experience

Quantitative Research Intern *Jun–Aug 2026*

Stealth Systematic Macro Fund; reference available on request *Hong Kong*

- Built a Bayesian regime estimator for global equity indices and adopted it into the fund's research stack. Monthly ranking strategies run on it lifted Sharpe by 0.05–0.15 versus the cross-section, cutting participation by up to 75%.
- Developed co-activity and flow measures tracking probability mass between regimes, isolating churn conditions.

Projects

Llama-Style Language Model *PyTorch*

- Implemented a 25M-parameter Llama-style transformer model in PyTorch, including BPE, rotary embeddings, grouped-query attention, RMSNorm, SwiGLU, and AdamW; trained on a custom 212M-token literary and philosophical corpus, with controlled upsampling experiments across selected authors.

Differentiable Kripke Semantics for Modal Model Search *PyTorch, NumPy*

- Implemented differentiable fuzzy Kripke semantics and cast modal satisfiability as gradient descent on the accessibility relation, posed as a minimax against an adversarial valuation. Developing a framework that uses modal constraints as a regulariser for RL policies.

Bitewise Labs — Bites Suite *Flutter, Swift*

- Founded an iOS studio and shipped three cultural-education apps (Literature, Art, and Architecture Bites): about 1,800 downloads and consistent revenue, across 2,323 quotations, 985 artworks, and 641 buildings. Built a short-form video pipeline.

Finite-Model Theory / FO² Decision Problem — Invited Seminar Presentation

- Researched, prepared and delivered a talk to faculty and members of the Australasian logic seminar on the decision problem for the binary fragment of first-order logic.

Paraconsistent Logic, Monash University *Lean 4, Isabelle*

- Formalized components of Weber's paraconsistent logic and set theory in Isabelle and Lean 4 whilst in high school. Supervised by professor David Ripley at Monash University.

Awards

Chancellor's Scholarship, the highest undergraduate award at the University of Sydney · **UWC Global Academy,** fully funded selective residential program at the University of Chicago · **Canadian Senior Maths Contest,** International Honour Roll, global top 1% · **SASMO Gold,** Global top 200. · **AMC10,** AIME Qualifier, global top 2.5%. · **AIME:** Scored 8/15.

Skills and Interests

Machine learning: PyTorch, NumPy, pandas · **Languages:** Python, SQL, LaTeX · **Also:** Flutter, Lean 4.

Interests: Music production (composed, produced, and released 5 albums with 40,000 streams on major streaming platforms; remix favourite songs on YouTube); golf; poker.